Business Sales Analysis

Retail transaction analysis | Python • Pandas • Matplotlib

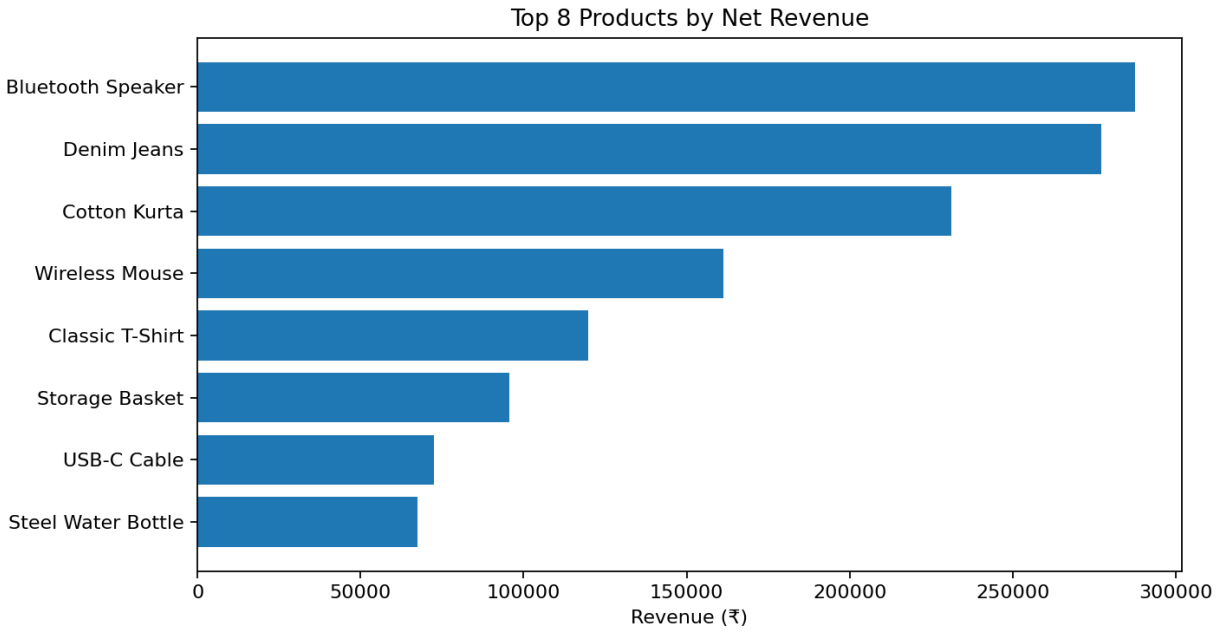
Dataset notice: This report analyzes a synthetic practice dataset created for portfolio demonstration. It is not real client data.

Metric	Result
Raw rows	1,258
Rows after cleaning	1,223
Rows removed during cleaning	35
Net revenue	■1,515,058.39
Units sold	2,492
Unique order IDs	1,223
Revenue per unique order ID	■1,238.80

1. Data preparation

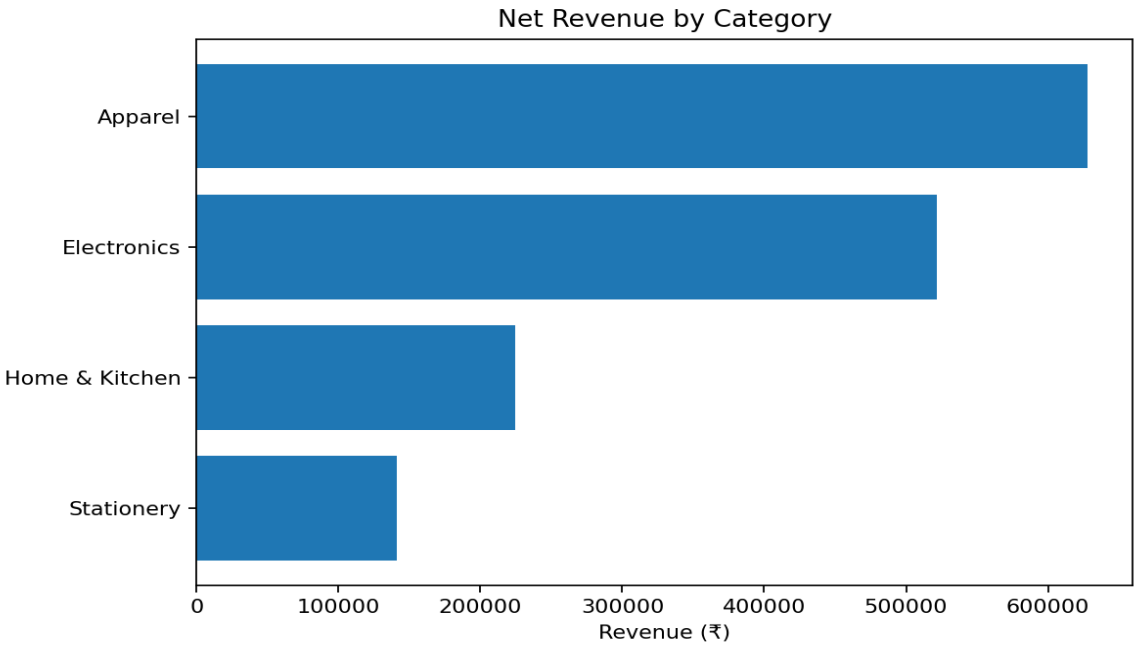
The raw dataset contained 1,258 rows and 8 exact duplicate rows before cleaning. The workflow standardized text fields, parsed dates and numeric columns, removed exact duplicates, excluded rows missing critical fields, and removed rows with non-positive quantity or unit price. Missing discounts were treated as zero; missing region/category labels were marked as Unknown. The final analysis uses 1,223 rows. These rules should be agreed with a real client before production use.

2. Product performance

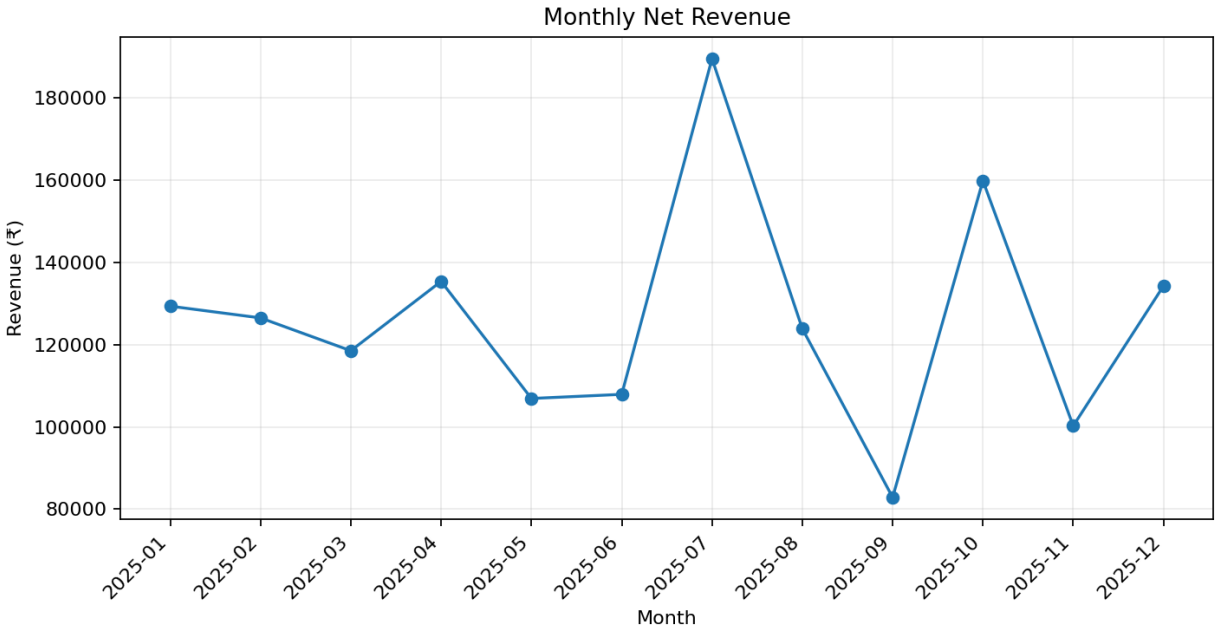


The highest-revenue product in this dataset is **Bluetooth Speaker**, contributing ■287,425.75. The top three products together contribute 52.5% of recorded net revenue. This suggests reviewing stock availability and product visibility for leading items, while also checking margins before making inventory decisions.

3. Category and monthly trends



The leading category is **Apparel** with ■628,209.96 net revenue. Monthly revenue peaks in **2025-07** and is lowest in **2025-09**. These are descriptive findings; the synthetic data cannot establish why the pattern occurred.



4. Practical recommendations

• **1. Protect availability of best sellers.** Review stock levels and stockout history for Bluetooth Speaker; compare sales volume and gross margin before increasing inventory.

• **2. Test category-level merchandising.** The Apparel category leads by revenue. Try a small, measurable placement or bundle experiment and compare conversion and margin.

• **3. Investigate the peak month.** Examine inventory, promotions, price changes, and marketing activity around 2025-07; repeat only what can be supported by evidence.

• **4. Diagnose the weakest month.** Check stockouts, seasonality, campaigns, and data completeness for 2025-09 before cutting spend or assuming demand is weak.

• **5. Use regional metrics with cost context.** South has the highest recorded regional revenue. Compare fulfillment costs, returns, and customer acquisition costs before shifting budget.

5. Limitations

Revenue is not profit and excludes product costs, shipping, tax, refunds, and operating expenses. The dataset is synthetic and intentionally messy. Order IDs are treated as transaction identifiers; real datasets may require line-item aggregation. Monthly patterns do not prove causation. Validate the workflow and definitions with the source system before using results for business decisions.

Reproducibility: Run business_sales_analysis.ipynb in the same folder as messy_retail_sales.csv. The notebook regenerates tables and charts from the CSV.